

Education

Madison, SD	Dakota State University	Fall 2023 – Spring 2027
• Bachelor of Science in Computer Science and Cyber Operations, GPA: 3.96 • Dean's List, Honors Program, DSU Presidential Champion scholarship • Coursework: Data Structures, Parallel Computing, Networking, Object Oriented Programming, Software Security, Assembly Language, Software Exploitation, Applications of Deep Learning		

Research Experience

Quantum Security Researcher	Dakota State University	Summer 2024
Multi-level Post Quantum Encryption for Images with Quantum Fourier Transform • Designed novel multi-level post quantum encryption algorithm combining QFT, AES, and LWE cryptography, achieving PSNR >40 dB and SSIM 0.95-0.98 while maintaining encryption times under 2.4s for images • Published peer reviewed research in IEEE Xplore		
Quantum AI Researcher	Dakota State University	Spring 2026
A Novel Hybrid Quantum-Classical Path Optimization for Methane Detection Using Remote Quantum Intensity Prediction Models • Built physics-aware hybrid quantum-classical UAV methane detection framework using RAMS/HYSPLIT plume simulations, environmental feature engineering, and PCA-based QIF • Published in IEEE Access. Demonstrated about 50% lower energy use than classical grid-routing baselines while maintaining 21 target coverage.		

Technical Experience

Projects
• Zig Data Structures & Algorithms Library: Implemented 10 data structures and 12 algorithms for systems programmers learning Zig • Neural Network Library in Zig: Built configurable neural networks with matrix operations, random weight initialization, and ReLU/sigmoid activations • Quantum Gaussian Process Regression in Python: Compared quantum and classical GPR for methane leak concentration prediction with tests and visual outputs • Monte Carlo Drug Simulation in C: Used OpenMP to model drug addiction spread across peer-influenced populations • Tracedog: Built TypeScript causal intelligence platform for OpenTelemetry trace analysis and LLM-narrated root-cause reports • Research Article Discussion Platform in Python: Built DOI-organized real-time chat rooms for academic paper discussion

Work Experience

STEM Academic Tutor	Dakota State University	Fall 2024 – Spring 2025
• Mentored GrowStem scholarship recipients in mathematics and programming fundamentals • Guided students through initial phases of undergraduate research project development		

Languages and Technologies

• Languages: Zig (proficient), C (proficient), Python (proficient), SQL (proficient), JavaScript (proficient), TypeScript (proficient), Java (intermediate), C# (intermediate), C++ (prior experience) • Technologies: Linux, Vim, Git, SQLite, PostgreSQL, OpenMP
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